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9
Future Direction in Sign Language Recognition: A Review

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Abstract

The obstacles that deaf and mute people face are universal, in that they are unable to convey or socialize with others in the same way normal people do. A sense of equity and inclusion is required. They are not regarded as a regular member of the group; rather, they are regarded as a unique individual. Thus, the difficulty they face is universal, but the sign language they use is not. Presently, there are apparently over 300 distinct sign languages incorporated worldwide. Every sign language is unique in

its own way. Sign language recognition was created to help normal people understand sign language, which deaf and mute people use to express their feelings and emotions through gestures such as hand gestures and facial expressions. Because they cannot communicate or interact socially with others in an identical way as hearing and speaking persons do, deaf and mute people encounter universal challenges. They are viewed as an unusual individual rather than a normal group member. Because of this, their challenge is shared by everyone, but their sign language is not. In the modern world, about 300 distinct sign languages are in use. Each sign language has its distinctive features. Deaf and mute persons use sign language to communicate their sentiments and emotions through hand and facial gestures. Sign language recognition (SLR) was developed to assist hearing and sighted people in understanding sign language to let them feel inclusive. Several creators across worldwide have created numerous sign language systems, for users as a whole, they are neither cost-effective nor versatile though.

Keywords**:** Connection, motion, facial expressiveness, hand motion recognition, classification

9.1 Introduction

Interaction using hand, facial expression, and movement of the body is called sign language. The use of gestures is advantageous for those who are considered hard of listening or deaf. The technique of converting the signals into sentences or characters of officially recognized languages is known as SLR. The disparity in communication that is present between the population of people who are hearing and speech impaired worldwide and those who are not can thus be reduced with the use of a technique or model that interprets words from sign language. Many people struggle with speech, hearing, or both. Hearing loss corresponds to the partial or total incapacity of one or both ears to perceive sound. Conversely, being silent is a disability that prevents a person from communicating and leaves those who experience it.

Sign language is a form of perceptual-spatial dialect that relies on placement and visual cues, including hand and finger shapes, hand location,

and orientation, and arm and motion of the body. Together, these elements serve to communicate the idea's meaning. The articulation point consists of the fingers, wrist, elbow, and shoulder joints; hand structure consists of fingers bending, movement of fingers, and bending of the palm; movement type consists of forward, backward, left, and right; hand direction consists of (palm up, palm down, palm inward, and sloping of palm; and gaze (happy facial, angry facial, movement of lips, and head shaking) represent the important components of sign language, which automated systems can use to recognize sign language, as mentioned in Figure 9.1 [1, 2].

The identification of gestures using the hands for human-computer interaction [3] is now a topic of ongoing research in ML and computer vision. One of the foremost objectives of this technology is to innovate systems capable of recognizing and interpreting distinct gestures to convey information or operate a device. However, it is necessary to consider gestures. Spatiotemporal is the phenomenon under investigation represented by a modeling approach that incorporates both spatial and temporal dimensions [4]. Specifically, the actions performed by a hand are being examined within the both space and time framework. Posture refers to the fixed arrangement of the hand, while a gesture is a dynamic movement made by the hand. The hand exhibits a dynamic range of movement. There exist essentially two sorts. There are two primary methodologies Vision-based approaches are utilized in the context of movement of hands identification and data glove approaches. The study focuses on the Sustainable development of a vision-based system that has the knack of recognizing signs in real time. The rationale behind selecting a system that is based on a certain framework or approach is multi-faced. The concept of vision pertains to its ability to offer a more streamlined and efficient approach.

Analysis of the identification of hand gestures for interactions among humans and computers is now underway in the fields of computer vision and ML. The creation of systems is one of this technology's main uses capable of recognizing and interpreting distinct gestures for the purpose of conveying information or operating a device. However, it is necessary to consider gestures [5]. A modeling approach that incorporates both spatial

and temporal dimensions represents the phenomenon under investigation. Spatial and temporal characteristics are essential for precise identification in SLR. The placements and physical characteristics of both hands and fingertips inside the signing area are referred to as spatial features. Important spatial elements in sign language entail hand shape, hand positioning, orientation, mobility routes, and facial reactions; on the other hand, temporal features are related to the order and time of motions. Hand mobility dynamics, consecutive patterns, signal duration, and interval between signs are important temporal elements. Specifically, the actions performed by a hand are being examined within this framework. Posture refers to the fixed arrangement of the hand, while a gesture is a dynamic movement made by the hand. The hand exhibits a dynamic range of movement. There exist essentially two sorts. There are two primary methodologies employed regarding gestures made with hand identification, vision-based techniques and data glove approaches. The primary topic of this work is this study focuses on the development of a vision-based system having the ability to recognize signs in real time. The topic of inquiry pertains to the recognition of language. The rationale behind selecting a system that is based on a certain framework or approach is multi-faced. The concept of vision pertains to its ability to offer a more streamlined and efficient approach.

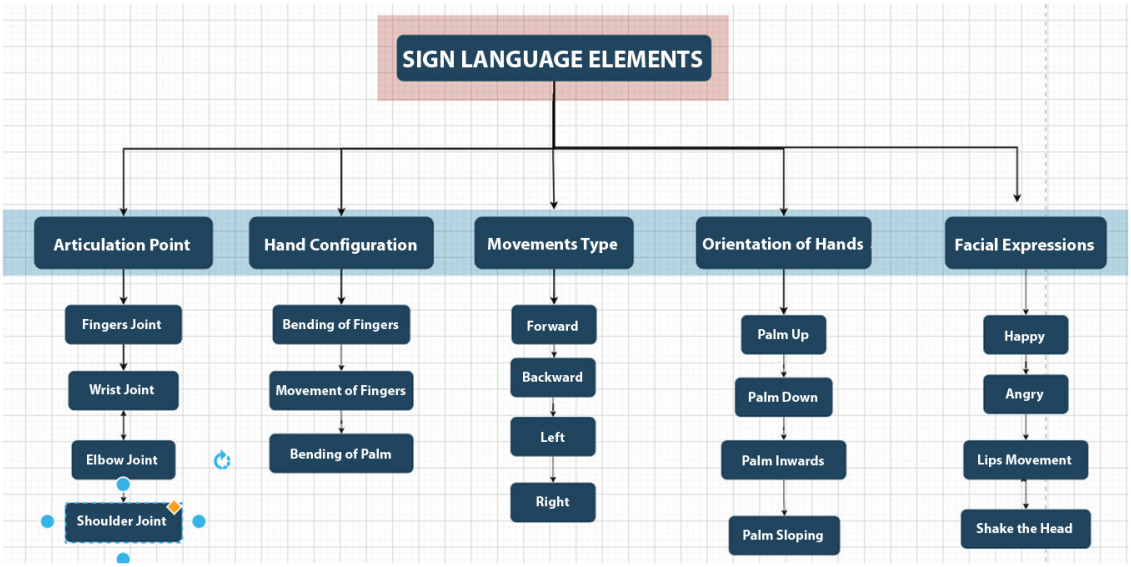


Figure 9.1 The essential elements of sign language gesture formation.

Communication tools play a significant role in the daily lives of individuals, facilitating interactions and connections with others. With the ongoing advancements in image and video processing techniques, there has

been a significant focus on exploring gesture recognition for human-machine interaction. Consequently, this technology has found extensive applications in various domains, including touch screens, video game consoles, augmented and virtual reality, health in resource-constrained settings, healthcare services, and SLR. Despite SL being the major mode of communication among individuals who are deaf, it has been noted that they encounter challenges when interacting with individuals who do not use sign language. Sign language is comprised of a lexicon of signs, similar to how spoken language is comprised of a lexicon of words.

9.2 Sign Languages Around the World

There is no one universal sign language like speaking the language. Around 300 different sign languages are currently used in the world which is not yet confirmed. Sign language among deaf and dumb communities has evolved similarly to the way spoken languages have evolved among the hearing community. Sign language includes French Sign Language (LSF), British Sign Language (BSL), American Sign Language (ASL), Australian Sign Language (Auslan), Chinese Sign Language (CSL), Arabic SL, Japanese SL, Danish SL, Thai SL, and many more, as we can see in Figure 9.2 [6]. Each language has distinct structural characteristics that distinguish it from the speaking language of the country which demonstrates that they are distinct languages on their own. It is interesting to observe that most nations with similar speaking languages do not have similar sign languages, for example, ASL, BSL, and Auslan. All these countries have almost the same speaking language, i.e., English but they vary in sign language.

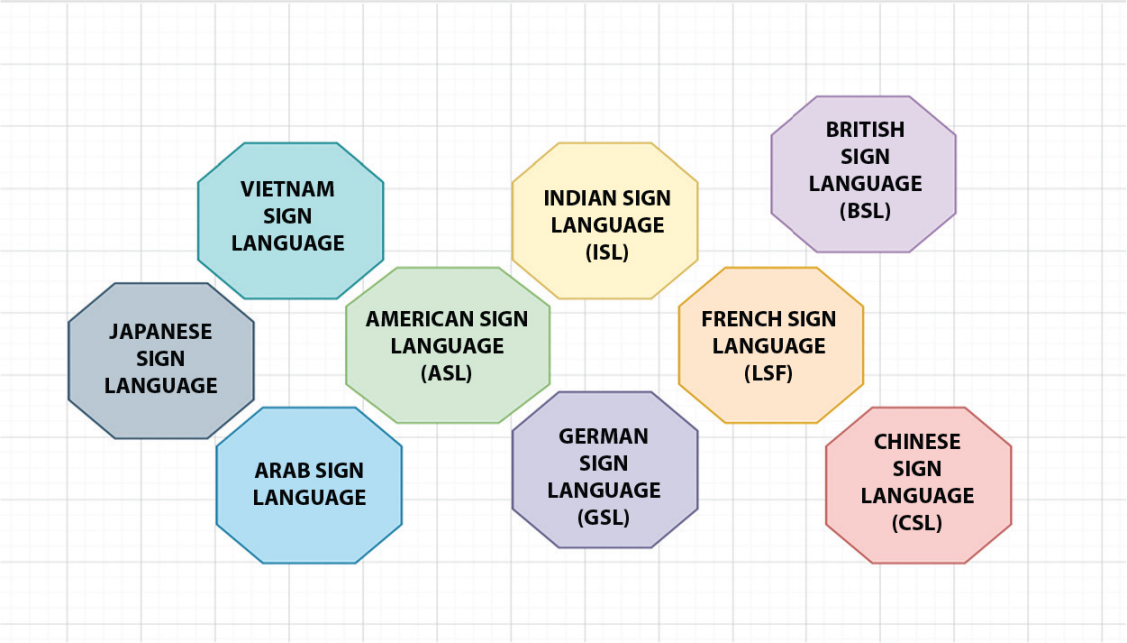


Figure 9.2 Sign languages around the world.

9.2.1 American Sign Language (ASL)

Within the Deaf and Dumb societies in the USA and many parts of Canada, ASL is one of the majority of widely used SL. It is essentially an offshoot of the antiquated LSF. In the USA, over 500,000 individuals speak ASL. There are particular grammar and syntactic rules unique to ASL. Like other languages, it is an evolving language that evolves and changes with time. There was no official use of sign language in the United States before 1814 when ASL history started. Without access to sign language as they grow up, deaf children may find it difficult to interact with others and take part fully in their communities. A priest named Dr. Thomas Hopkins Gallaudet, whose daughter was deaf, traveled to Europe in 1814 to study deaf teaching practices. After his return, he founded the American School of Deaf (ASD) in Hartford, Connecticut, and developed ASL as a result of a linguistic contact situation. Since he created ASL from LSF, thus there are some similarities between the two languages. Despite being widely used, ASL users are not properly counted. Reliable estimated that there are currently around 250,000 to 500,000 individuals who use ASL in the United States, many of whom are hearing persons or the offspring of deaf adults. As seen in Figure 9.3 [7, 8], the phonemic elements of ASL signs include the face, body, and hands.

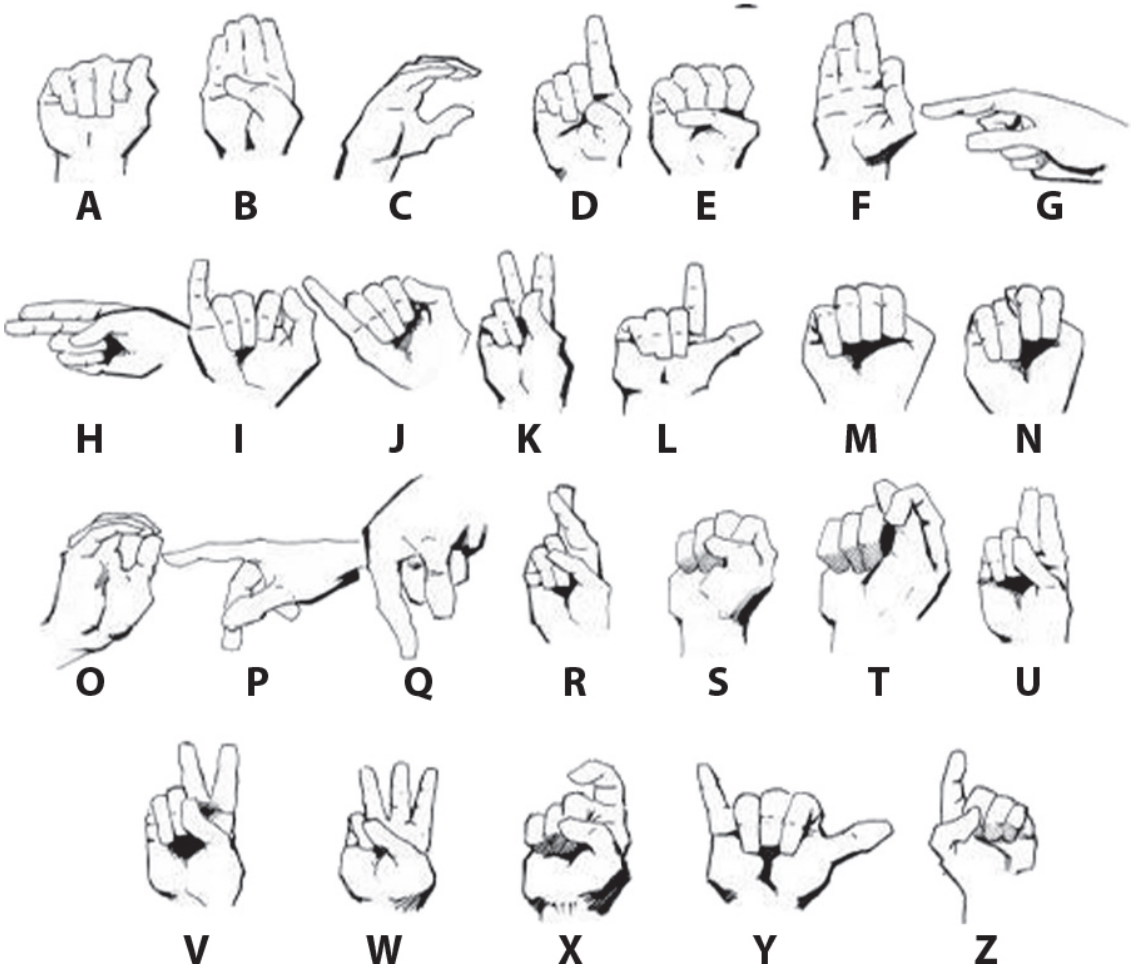


Figure 9.3 The standard ASL alphabets.

9.2.2 British Sign Language (BSL)

The primary SL spoken by deaf and mute individuals within the UK is BSL, a form of sign language. In the UK, BSL is the most widely used language for gestures. It gained official recognition as an indigenous language in its individuality in 2003 and as one of Britain's official languages in 2022. As seen in Figure 9.4 [9, 10], BSL primarily uses both hands for interpretation and communication. Because BSL has its grammatical structure and grammar, it is not reliant on spoken English and does not have any major similarities with it.

BSL was officially acknowledged as a recognized minority language by the UK government in 2003, after a significant struggle. This led to increased funding for the communication needs of the deaf and mute as well as more awareness of the language, which is now on par with comparable minority national languages like Welsh or Gaelic.

9.2.3 Chinese Sign Language (CSL)

Since the 1950s, the CSL has been developing. On September 23rd the IDSL which stands for International Day of Sign Languages marks the commencement of International Deaf Week from 23rd September to 29th September, there is a global emphasis on the full recognition of the linguistic rights of individuals with sharing disabilities. As a result, the theme for IDSL is “Sign Language Rights for ALL” [8].

After seven years of research, the Chinese State Language Commission has published national guidelines for sign language, according to a Beijing News story. It was released on July 1, 2018, on the nationwide list of common words for sign language, making it the first of its type in China. The most widely used reference books on CSL provide a list of 5,000 frequently used vocabulary. This is intended to enhance communication for people with disabilities across the nation as mentioned in Figure 9.5 [11].

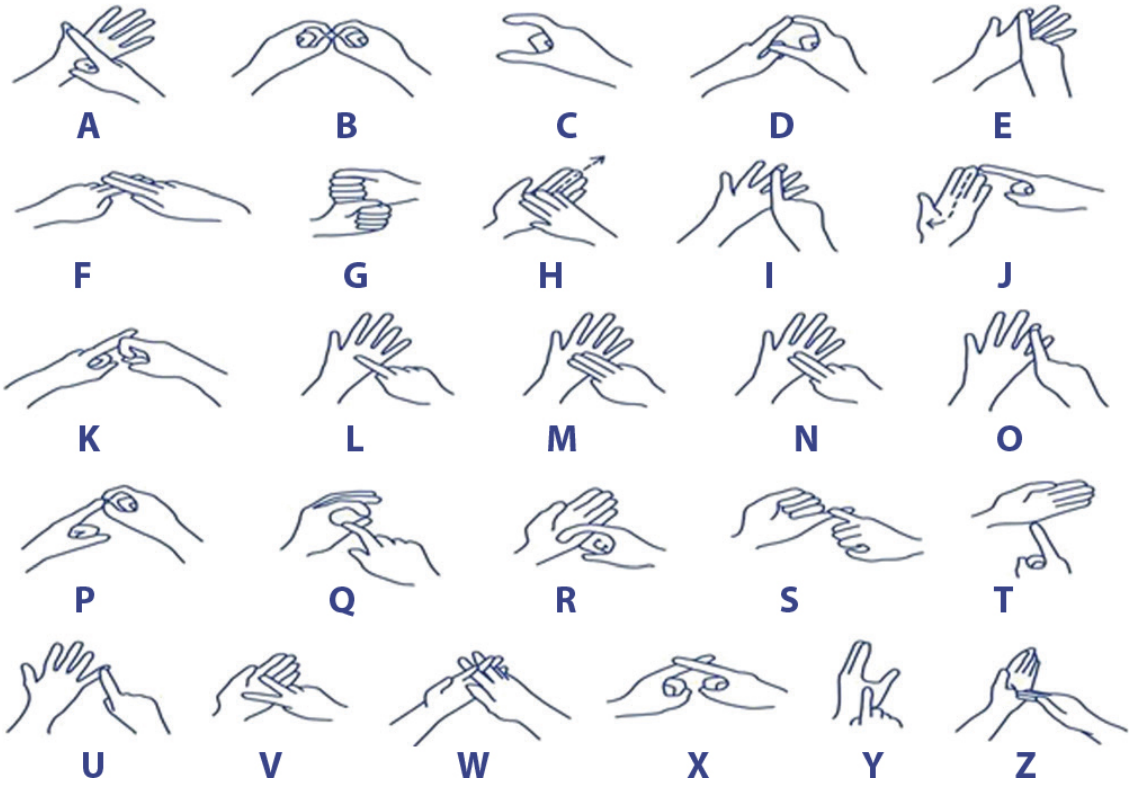


Figure 9.4 The standard BSL alphabets.

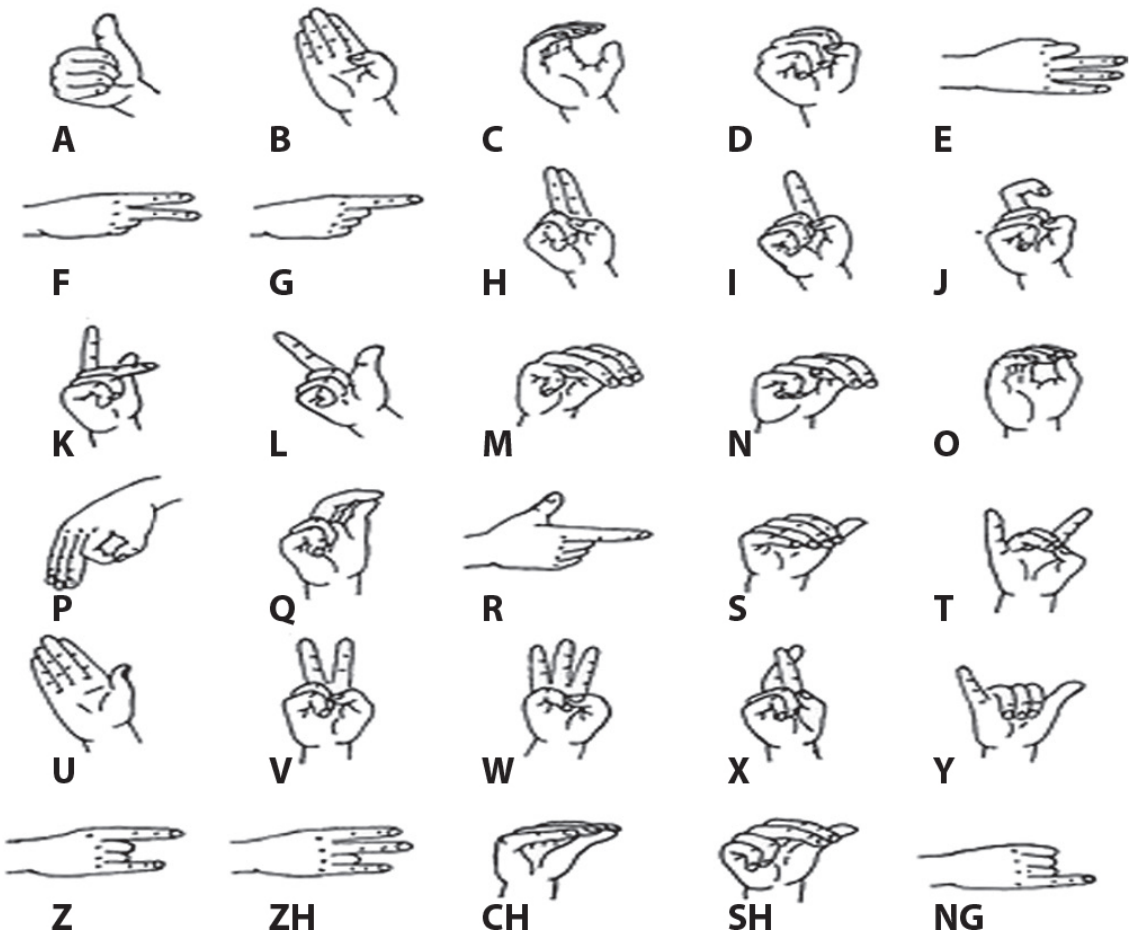


Figure 9.5 The standard CSL alphabets.

Based on recent data according to the China Disabled Persons Federation website China is home to 20.54 million deaf individuals, which accounts

for about 30% of the 72 million deaf persons in the globe reported by the World Federation of Deaf (WFD).

CSL has two primary dialects [9]: Northern CSL, which emerged from the Chefoo School of Deaf and is influenced by ASL, and Southern CSL, which is centered on Shanghai and influenced by FSL. The larger Chinese influence is in the northern CSL.

9.2.4 Australian Sign Language (Auslan)

In the 19th century, deaf people moved to Australia from Ireland, Scotland, and the United Kingdom, bringing their sign languages with them. Over time, Auslan has changed to acquire its unique characteristics. Australian deaf people developed Auslan so they could communicate with one another. Early in the 1980s, Trevor Johnston coined the term Auslan, which is an abbreviation for Auslan. Like other sign languages, Auslan differs greatly from English in terms of syntax and vocabulary as mentioned in Figure 9.6 [12]. It is an instinctual language that has developed on its own over time; no one person can be held responsible for its expansion. In 1987 and 1991, the Australian government announced legislative changes designating the Deaf Community to prefer Auslan as their “Community language other than English.”

9.2.5 Indian Sign Language (ISL)

The Ramakrishna Mission and CBM International, Germany, started working on a very special Initiative in January 1999 [12] that is standardizing Indian Sign Language as we can see different Indian Signs in Figure 9.7 [11, 13]. In India, November 24, 2001, came to be recognized as a significant day for those who have hearing loss. The first “Indian Sign Language Dictionary” was published during a formal event and was a sincere attempt to offer a Uniform Sign Language Code throughout India. It was built on years of research and documented over 2,500 signs from 42 locations in 12 states. The Indian Sign Language (ISL) teaching and research institute was supported by The Indian Deaf Community in the 2000s. The 11th Five-Year Plan (2207-2012) recognized that the requirements of those with hearing impairments had been largely disregarded

and proposed the establishment of a center for sign language research and training in order to advance the field, as well as to train instructors and interpreters [14].

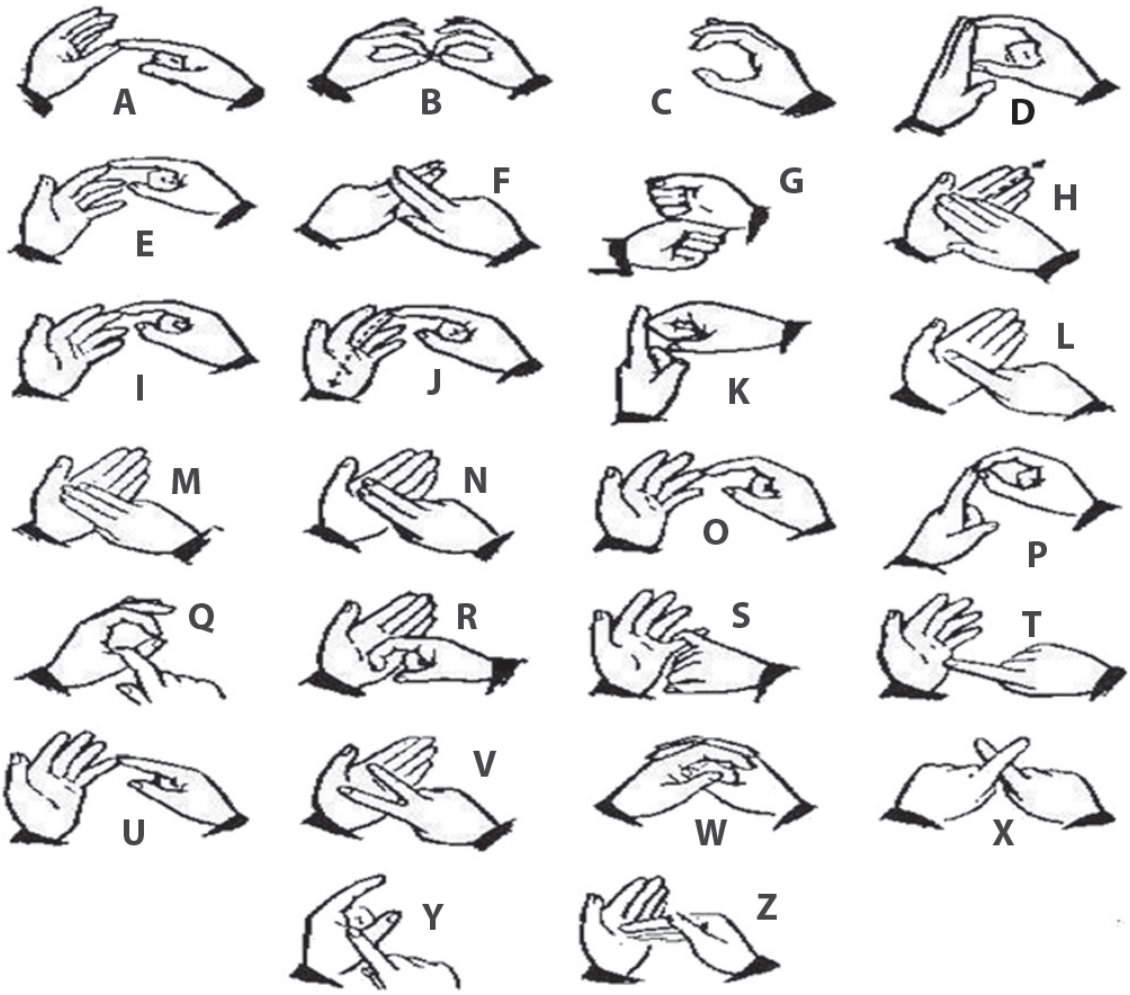


Figure 9.6 The standard Auslan alphabets.

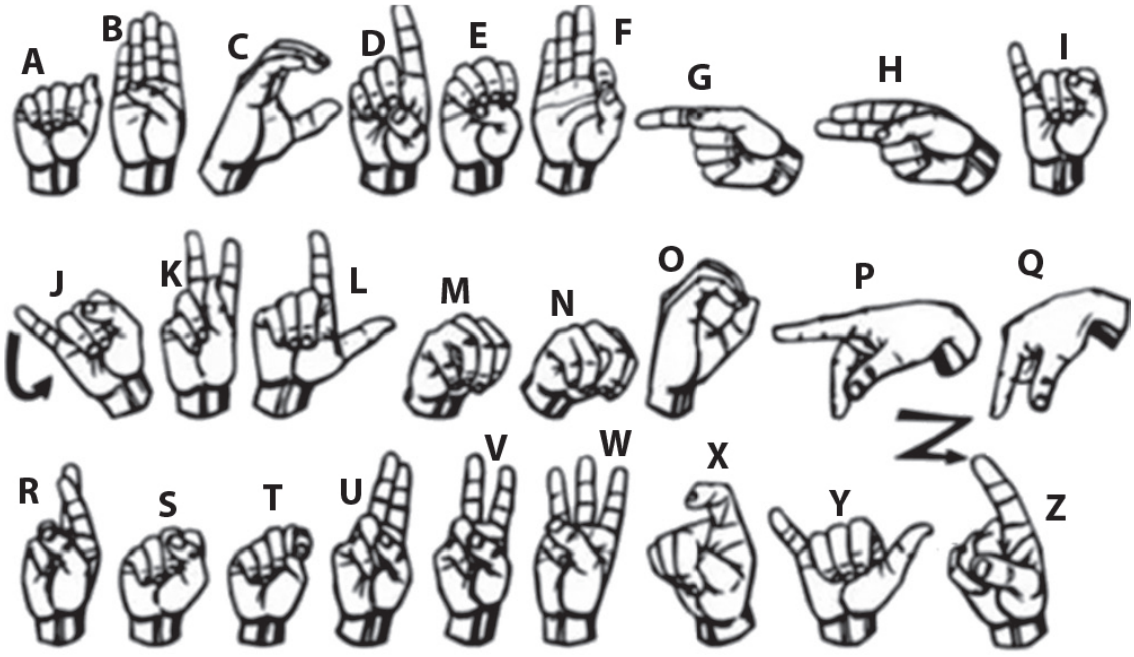


Figure 9.7 The standard ISL alphabets.

9.3 Sign Language Linguistics

Sign language is used by deaf and mute people to communicate inside and outside of their respective communities in everyday life. They use hand gestures to convey their feelings or to communicate with other people. The sign language recognition (SLR) objective is to recognize learned hand signals and to keep going until the corresponding hand movements or gestures are translated into text or speech. Hand gestures are classified as static and dynamic. Although dynamic hand gesture recognition is more complex than static gesture recognition, both recognitions are significant to humans.

Although sign language recognition research has advanced in recent years, commercially, it is still in its infancy phase. As a result of the need for experienced interpreters, all commercial interpreting basic services are now human resources based as well as expensive.

The creation of algorithms and procedures that correctly identify several produced signs and understand their meaning is another goal of sign language recognition (SLR). Many SLR techniques are incorrectly labeled as gesture recognition (GR). This has led to a particular emphasis on the current research on the identification of the best characteristics and classification methods so that one characteristic can be precisely described from this set of possible indications. A functioning SLR system can enable a non-signer to communicate with a silent person without the need for an interpreter. It may be used for writing or speaking, providing the silent person with more freedom.

There is much more to sign language than merely a set of precise hand gestures. The difficulty with sign languages is that they use multiple channels to communicate meaning simultaneously. The phonology, morphology, syntax, and grammar of sign languages are well-structured and different from those of spoken languages. Multiple simultaneous body moments in both spatial and temporal space are used in sign language. Due to the presence of many context-affecting factors such as the use of facial expressions, sign languages have different linguistic properties from spoken language [15].

Manual and **Non-Manual** elements are the two primary divisions of sign language. These components combine to help sign languages transmit meaning. Hand motions, or the use of hand forms and motions to express information, are considered manual aspects. In contrast, non-manual elements like body posture or emotions displayed on the face can either be a component of a manual sign or alter its meaning [16].

In Manual Elements, the first component is **Signs** (signs are the primary lexical unit in sign language). It includes hand shapes, actions, body placements, and palm orientations. Individual words or concepts can be simply represented by signs. Another part of the Manual Element is **Finger Spelling** (it is basically a practice of using hand signs to represent individual letters or in some situations, entire words). It is used to spell out names, technical terminology, and words without a specific sign. Similarly, the **Numerical Sign** is also a part of the Manual Element (in these many sign languages include their own signs for numbers, allowing numerical information to be expressed.)

In Non-Manual Elements, the first component is **Facial Expressions** (when learning sign language, expressions of the face are crucial). They convey a range of grammatical and emotional information. A sign's meaning or intensity can be altered by changing its facial expression. The second component of the Non-Manual Element is **Head Movement** (head movement such as nodding or shaking can express questions, negations, or affirmations). The third component is **Body Posture** (the signer's posture and movement can also alter the meaning of signs or reveal information about the spatial arrangement of things or persons). The fourth component is **Eye Glance** (it is used for establishing referents in the signing space, indicates who is speaking, and conveys information about individuals or objects). Another component is **Mouth Motions.**

Simply, sign language combines non-manual and manual sections to create a complex and expressive mode of communication.

9.4 Motivation

The SLR's objective is to establish an interaction between humans and computer tools that can understand sign language more effectively than previous works on various techniques like image preprocessing, sensor data processing, and motion detection. The establishment of algorithms is SLR's main goal and methodologies that can accurately discern a series of generated signs and comprehend their intended significance. Numerous methodologies in the field of SLR exhibit a flawed perspective by erroneously considering the challenge of gesture recognition. Previous studies have mostly concentrated on the identification of appropriate features and categorization methods to precisely identify a particular indication from a collection of possible indications. That being said, sign language is greater than simply a list of well-defined movements.

Based on the data published by the World Health Organization [17], the global prevalence of hearing impairment in 2005 was estimated to be over 278 million individuals. A decade later, the figure experienced a significant increase to 360 million, reflecting a growth of almost 14%. Subsequently, the numerical value has been experiencing exponential growth. According to the most recent data published by the WHO [18, 19], the prevalence of deafness in 2019 was estimated to affect approximately 466 million individuals, constituting approximately 5% of global citizens. Among this affected population, 432 million individuals, accounting for approximately 83%, were adults, while the remaining 34 million individuals, comprising approximately 17% were children.

According to the WHO [18, 19], it was also projected that the aforementioned figure will experience a twofold increase, reaching around 900 million individuals, by the year 2050. In light of the rapid growth of the deaf-mute population, the impediment to communication must be addressed as it has a substantial negative influence on the social interactions and quality of life for those who have speech and hearing impairments.

9.5 Objective

Earlier people with special needs faced difficulty in cooperating with ordinary people, which became quite difficult for them to complete daily tasks as ordinary people can do easily without any difficulty. Hence, it was becoming a serious issue for them as society started treating them as different people not considering them as one of them thus, people with special needs started getting neglected in society.

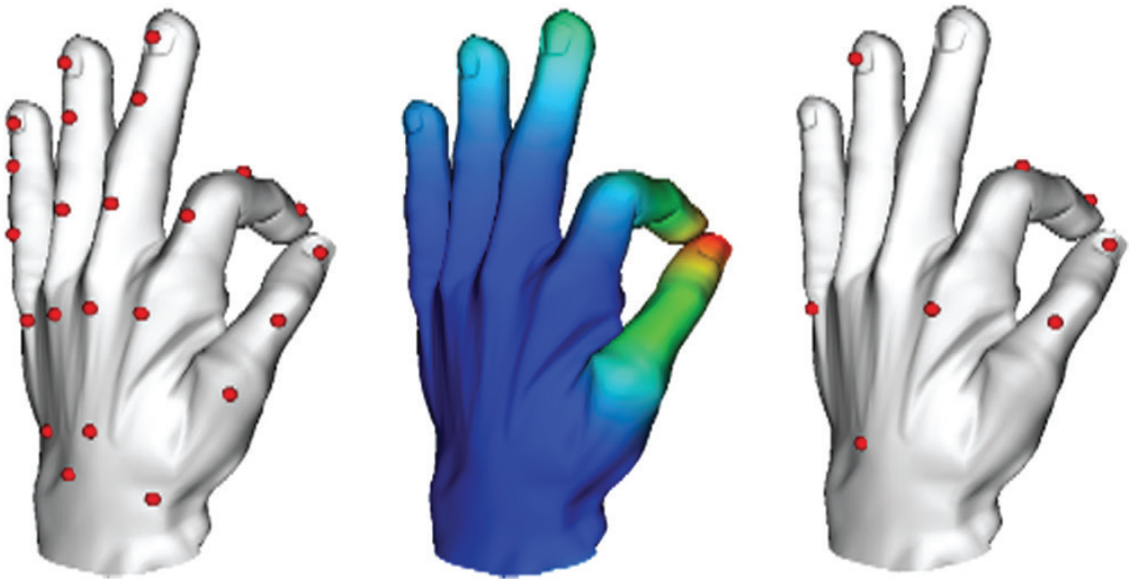


Figure 9.8 Analyzing hand moments [21].

- Our objective is to help the special need with the help of algorithms and neural networks [20]. We analyzed and studied various hand gestures used by special needs people for communication similar to those mentioned in Figure 9.8 [21] using image preprocessed datasets [9].
- Another objective is the collection of data to understand sign language and its existing related issues.
- Similarly, our third objective will be to employ relevant datasets to apply machine learning to the recognition and interpretation of data.

9.6 Related Work

How computers and humans communicate has evolved in tandem with the advancements in information technology. Much work has been carried out in this area to assist deaf and hearing individuals more communicating effectively. Since sign language is made up of movements, any at-

tempt to comprehend sign language falls flat under the responsibility of human-machine interaction.

In one of the studies [5], R. Rumana, R. S Rani, and Mrs. R. Prema designed an SLR realtime prototype to accurately identify and interpret ASL gestures as depicted in alphabets. The prototype's main goal was to evaluate the effectiveness of a vision-based mechanism intended for the recognition of sign languages. Additionally, the prototype aimed to evaluate and identify hand characteristics that could be effectively employed in machine learning algorithms, hence enabling their integration into real-time recognition of sign language systems. Its purpose is to enhance communication efficacy between individuals who are deaf and mute, as well as those who possess normal speech and hearing abilities.

In similar another study [22], Amrutha K. and Prabu P. presented an isolated recognition model created for a rudimentary SLR system. The mechanism is based on solitary hand motion detection and identification using vision. A controlled setting was used to evaluate the ML-based SLR model with the help of four candidates. A convex hull was used for feature extraction, while KNN was used for classification. The model had a 65% accuracy rate.

In another study [23], A. Wadhawan and P. Kumar focused on reliable modeling of static gestures using CNNs based on deep learning for the recognition of sign language. 35,000 sign photos of 100 fixed signs in total were accumulated from numerous individuals for this study. The suggested system's effectiveness is assessed using about 50 CNN models. The suggested approach has been reported to attain the greatest proficiency in the training of 99.90% on grayscale photos and 99.72% on colorful images, accordingly. The results are also evaluated based on several optimizers. F-score, recall, and precision have all been used to assess the suggested system's performance.

In "An Automatic Arabic SLR System" [24] by Nada B. Ibrahim *et al.*, the four primary phases of the suggested system are feature extraction, classification, tracking, and hand segmentation. Hands are segmented using an interactive skin detection system based on face color tone, and hands are also identified and tracked using a skin-blob tracking approach.

According to experimental findings, the suggested system has a 97% signer-independent recognition rate. Furthermore, with an enhancement of 2.57%, the suggested occlusion resolution strategy can perform better than alternative approaches by precisely defining the hands and head positions.

In another similar research on the same study [25], Rybach D. *et al.* on "Speech Recognition Techniques for SLR systems" leveraged the knowledge gained from the automatic speech recognition (ASR) research by starting with a voice recognition system with a big vocabulary. The created system is able to identify continuous sign language sentences without the speaker's assistance. Standard video cameras provide the functionalities that are needed no specialized data-collecting equipment is required. We pay particular attention to the use of language models (LMs) and pronunciation in sign language, as well as characteristic and model combination approaches in ASR. These methods apply to a wide range of detection of sign language applications as well as several video analysis issues where time context plays a significant role. They obtained a 17% WER on a benchmark database that is accessible to the public and has 201 statements and 3 signers.

In another work [26], the components of the SLR system include image acquisition, segmentation, extraction of features, and recognition or classification. They presented a survey of feature extraction and segmentation methods for Indian SLR systems in this study.

The recent study [27] showcased the dataset that contains 71,400 images from two different sources that represent the Kurds' 34 SL and alphabets. A vast collection of genuine user photographs is used to assess the correctness of the proposed technique. This study presents a novel categorization model based on Kurdish Sign Language (KuSL). Additionally, the area surrounding the hand must be detected in a photograph with a complicated backdrop, which includes lightning, ambient, and imaging color changes of varied intensities. The compiled statistics show that the efficiency of the suggested methodology improves, with an average accuracy for training of 99.05 percent for both the prediction and classification.

Similarly, in another recent study [28], the researchers examined different approaches for creating synthetic samples using directed ASL data, but they finally demonstrated that using such data is not significantly better than initializing with ImageNet images. The authors, on the other hand, suggested an asynchronous learning paradigm in which it is demonstrated that model training gets better with the more natural ASL examples that are collected and supplemented with synthetic samples produced by a generative adversarial network that is aware of physics. The authors demonstrated how the suggested method improves upon conventional instruction with directed ASL data by 60%, allowing for the accurate detection of natural ASL in an everyday setting with a precision of 69% for 29 ASL signals.

Another study [29] suggested using this dataset to provide ASL sign language dictionary retrieval, in which a user can obtain matching signs from a dictionary by demonstrating a sign to their webcam. We demonstrate that by using our generalizable baselines, they can train supervised machine learning classification algorithms with our dataset and obtain competitive results on metrics related to dictionary retrieval, with a recall-at-10 of 91% and 63% accuracy when evaluated.

The findings of the authors' [30] investigation on the significance of non-manual aspects for recognition of sign language systems indicated that the accuracy of recognition was increased by around 4% when hand and facial key points were combined. The Arabic and Argentinean Sign Languages were used to test the suggested models. The suggested pose-based transformer obtained the maximum precision of 99.74% and 68.2% in signers independent and dependent modes, respectively using the KArSL-100 dataset.

9.7 Approaches

Taking into account that the deaf community uses gestures as their primary means of communication that are not deaf, the necessity for sign language translators (SLTs) arises. There exists a multitude of scholarly interventions that aim to address challenges associated with disabilities and mental health. These interventions are characterized by their system-

atic approach and can vary depending on the specific setting in which they are implemented. One of the significant interventions involves the utilization of SLR systems, which are employed to convert the elements of sign language into textual or spoken form. This facilitates communication with those who lack knowledge of these signs. The fundamental automatic SLT system employs two distinct methodologies [35]:

- Glove-Based Technique
- Vision-Based Approach

9.7.1 Glove-Based Technique

This particular category necessitates the utilization of either a sensor-equipped glove or a glove of a distinct hue by the signers. The segmentation procedure can be facilitated by wearing gloves, hence simplifying the operation. The utilization of a certain category of gloves equipped with diverse sensors, such as flex sensors, Arduino, and accelerometers, presents an alternate method for obtaining gesture–related data. The aforementioned sensors are employed for the purpose of quantifying the flexion angles of the fingers, the separation between adjacent fingers, as well as the spatial orientation of the wrist, encompassing roll, pitch, and yaw. One notable benefit of glove-based systems, in comparison to vision-based systems, is their ability to directly transmit pertinent and necessary data in the form of measurements of voltage transmitted to computer equipment, as shown in Figure 9.9 [10, 36]. Consequently, this obviates the necessity of converting unprocessed information into meaningful values through additional processing. One limitation of this technique is that the individual performing the signing must wear both the sensor hardware and the glove while using the system [37, 38].

The glove-based device for recognition consists of three essential hardware aspects, i.e., input, computation, and output, as shown in Figure 9.10. Within the input unit sensors, touch sensors and an IMU, which stands for inertial measurement unit, are used to collect data. A computation or processing unit is a further hardware component that uses a microcontroller board to handle data collected from sensors. The output unit is the final hardware element that shows the categorized motions on liquid crystal displays, laptops, microphones, and cellular devices.

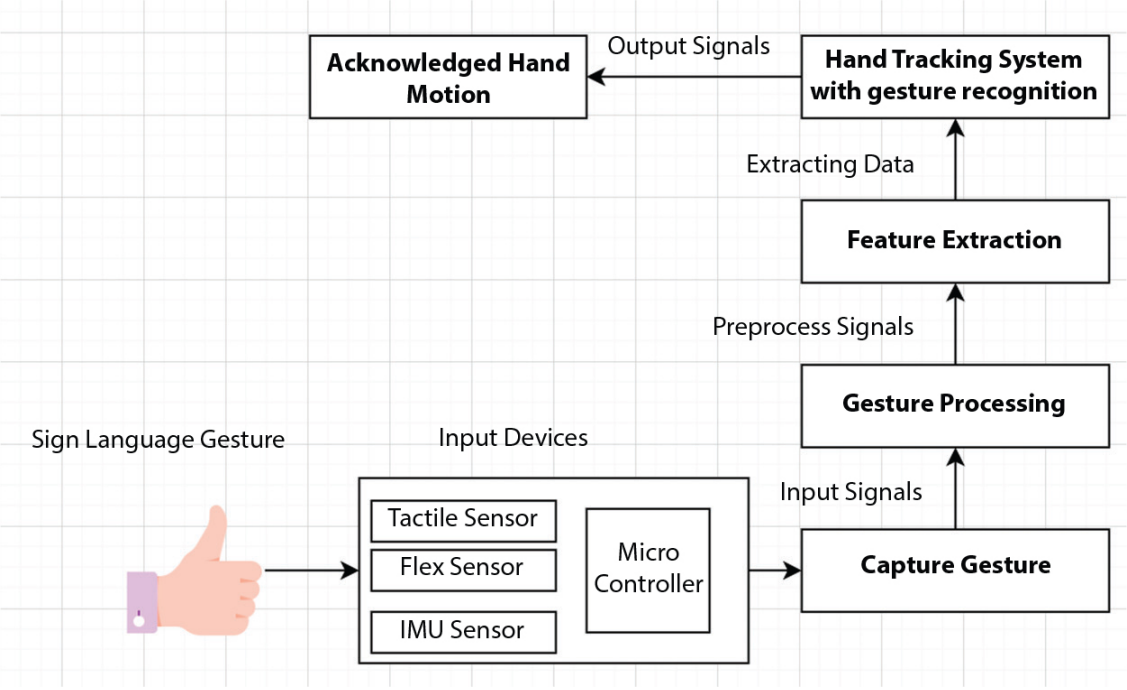


Figure 9.9 Block diagram for sign language recognition using glove-based technique.

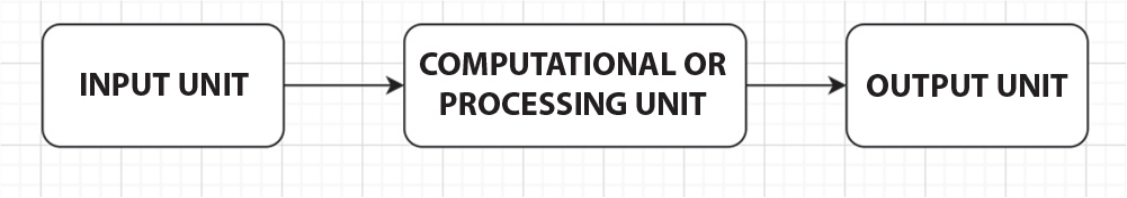


Figure 9.10 Hardware components for glove-based technique.

9.7.2 Vision-Based Technique

Vision-based techniques utilize image processing algorithms, as mentioned in Figure 9.11 [36], to detect and monitor hand signs and facial emotions exhibited by the signer. Cameras serve as the principal instruments in vision-based systems for acquiring the requisite input data. One primary benefit associated with the utilization of a camera is the elimination of the requirement for sensors in sensitive gloves, therefore leading to a decrease in the system's aggregate building costs. Cameras are relatively inexpensive, and a majority of laptops are equipped with high-quality cameras to mitigate the blurring effects commonly associated with web cameras. Nevertheless, despite the presence of high-quality cameras in the majority of smartphones, there exist several challenges like the constrained area of view of the capturing device, the substantial computational expenses involved, and the necessity of employing several cameras to achieve reliable outcomes. The aforementioned flaws are in-

herent to the current system. Consequently, the system as a whole is rendered useless for creating real-time recognition applications [33, 39].

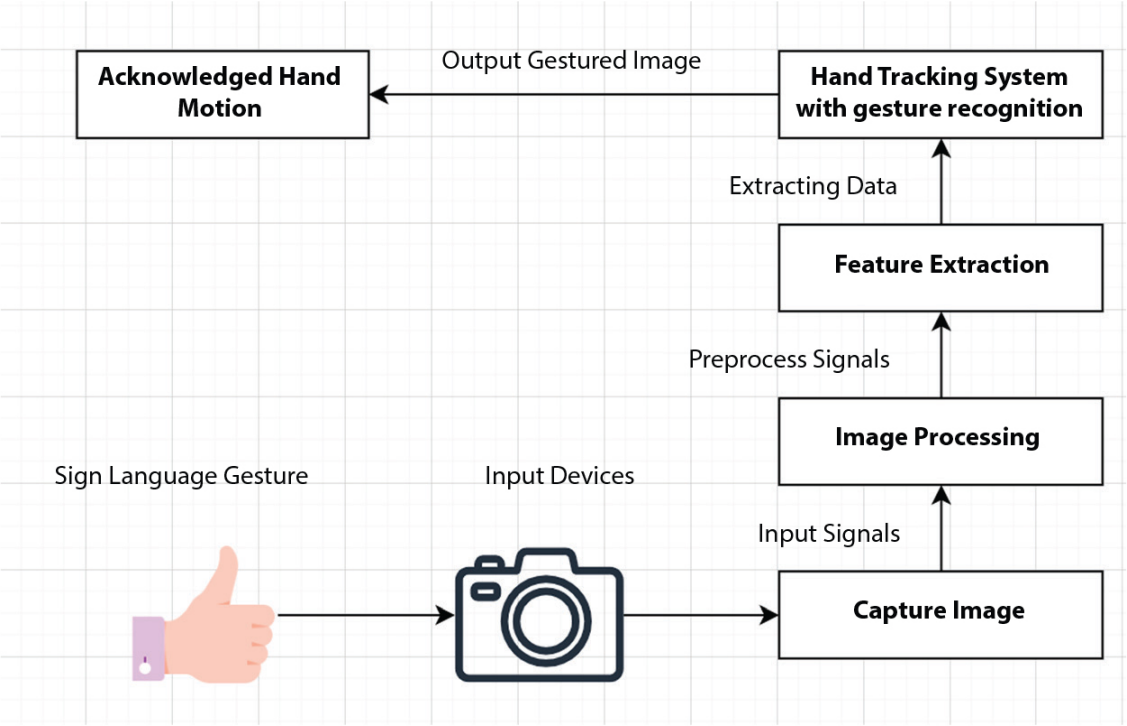


Figure 9.11 Block diagram for SLR using vision-based technique.

In simple words, the methodology involves the initial action involving the utilization of gloves. The gloves process integrated sensors and are used by individuals proficient in sign language to detect and analyze hand posture. However, it should be noted that these gloves come with a significant price tag. The visual technique is considered the most appropriate and commonly employed method. In this scenario, a camera is employed to collect a visual representation of the signer, followed by the application of image processing techniques to facilitate the process of recognition. There exists a multitude of approaches for the visionbased approach [40].

9.8 Proposed Methodology

It can be challenging for non-deaf and non-mute persons to grasp the communication method used by deaf and mute individuals, sign language. Creating a method to close the communication gap between the deaf and mute community and the broader public is the stated difficulty in this situation [31]. A sign language recognition (SLR) system's goal is to accurately translate provided signs to text or written content. It is a multi-

disciplinary research field that integrates psychology, computer vision, pattern recognition, and natural language processing, as shown in Figure 9.12 [32].

Classification, preprocessing, and extraction of features are crucial procedures in the sign language recognition system.

9.8.1 Training Dataset/Data Acquisition

A collection of information that is input, also known as characteristics, and the associated output values, also known as target variables, make up a dataset. The majority of academics create their individual databases for training with data. Although gesture-based datasets are not always easily accessible, researchers capture the signer's data to create a dataset.

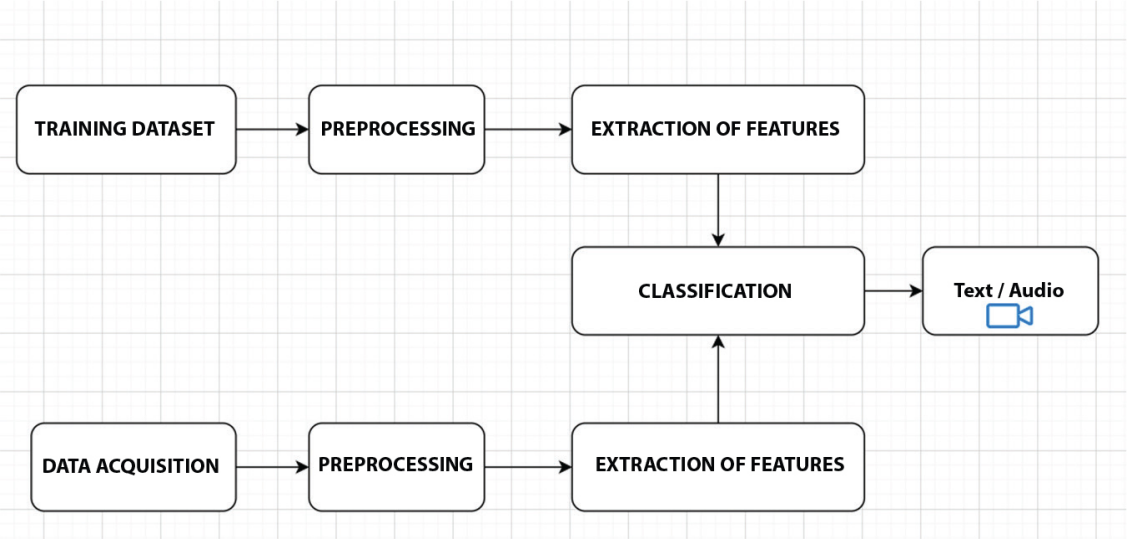


Figure 9.12 Basic block diagram for SLR.

One step in the SLR process is data acquisition, which is digitizing user input to make it suitable for classification algorithms. Depending on the tools being used, user input can be collected in a variety of ways, such as visually through recording devices or wearably through the installation of gauges in a glove. While those instruments offer several benefits, including affordability, interoperability with various software and hardware components, user detection, etc., they also have drawbacks, like obstruction, expense, usability difficulties, etc. The instruments used to gather user data differ based on the nature of the study.

9.8.2 Preprocessing

The ROI, i.e., Region of Interest, is taken out of the training images using preprocessing. It can be the hands only if it is hand motions are considered, or the face and hands if facial expressions are also included. They are employed to eliminate unnecessary noise and improve the clarity of input snapshots. To do so, we may use a resize, color correction, remove unwanted noise, or merge the original image's approaches. The use of appropriate preprocessing methods may substantially affect the accuracy of production from this process. The two primary subcategories of image preprocessing techniques are enhancing and restoring images. Image-enhancing methods are referred to by the acronyms HE, AHE, CLAHE, and logarithmic transformation. Image restoration uses a variety of filters, including Gaussian, median, mean, adaptive, and others [23, 33].

9.8.3 Feature Extraction

The extraction of the feature vectors acquired in this stage constitutes one of the sign language recognition (SLR) most crucial responsibilities. Since these features are input to a classifier. The feature extraction techniques should remain capable of detecting shapes with accuracy in all situations where an item is lightning, and position, and size changes during a video or image. For this purpose, the classifier is trained using a feature vector that was generated from any of the feature extraction methods.

9.8.4 Classification

After modeling and evaluating the input hand pictures or hand motion, the technique of classification is utilized to identify the motions and generate an audio note or text that corresponds to the correctly categorized sign [34]. Recognition is affected by choosing the right feature parameter and classification algorithm. Statistical approaches and gentle computational methods are used to identify each hand position. There are two types of ML algorithms for classifying, unsupervised and supervised. It is known as supervised machine learning when you train the computer to recognize patterns in input data, which may then be used to predict future data. The function is inferred from the labeled training data with

known training data in supervised machine learning. Whereas, unsupervised ML is used to infer results from data that do not contain tag responses. Because no labeled response is provided to the classifier, there is no reward or penalty weighting for which classes of data are expected to be discarded. Some of the most popular classifiers include SVM, KNN, and ANN.

9.9 Conclusion and Future Scope

The process of SLR has significant challenges due to the distinctiveness of each sign, which is specific to individual signers and encompasses a range of manual and non-manual elements. Numerous datasets exhibit imbalanced characteristics and possess limited vocabulary due to the high data requirements of deep learning techniques, which might pose challenges in acquiring data from individuals proficient in sign language. In summary, it is possible to conceptualize a framework that would have a positive impact on global society by effectively addressing the communication divide between two distinct populations. The proposed solution will effectively address the existing communication gap. Specialized algorithms and functions are employed to actualize this aspiration. The application utilizes a well-trained model to analyze the image and identify individual elements. It then compares these elements and generates the desired output. The objective of this study is to eliminate constraints inside the system and transform it into a bidirectional communication tool. Furthermore, the system might potentially be improved by transforming it into a mobile application, thereby facilitating convenient access for individuals within society.

The experiences of individuals categorized as hard of hearing or deaf might soon be significantly impacted by sign language recognition technology. Improving the written and voice outputs' correctness and reaction time constitutes one of the crucial tasks that has to be done in the future. Since SLT datasets remain small, it will be challenging to draw interferences from a small number of data instances. This is yet another pertinent subject for further investigation. A sizable dataset could hold vital information to facilitate more study and the exploration of creative model concepts. We may anticipate a day in the future when the deaf commu-

nity will face a great reduction of interaction obstacles with more advancements and greater understanding.

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